Real-Time Emotion Detection based on Physiological, Vocal, and Facial Modalities Using Deep Learning

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***Abstract*—Recognizing human emotions in real time has become increasingly relevant in areas such as mental wellness, smart systems, and responsive human-computer interaction. Emotion detection systems relying on a single input type like facial cues, speech tone, or bio signals often fail to deliver consistent performance in real life environments because of noise, changing contexts, and person specific variations. This study proposes a multimodal emotion detection system that fuses physiological signals (heart rate, SpO_2, body temperature), facial expressions, and speech features to enhance classification accuracy and robustness. Each modality is individually processed; facial data is analyzed using a CNN trained on FER-2013, vocal features are extracted via MFCCs and modeled with an LSTM, and physiological inputs are classified using a Random Forest algorithm. A weighted fusion technique combines predictions from all three modalities, with 60 % weightage to physiological data and 20 each to facial and vocal inputs. Data is acquired in real time using an ESP32 based system integrated with Firebase Realtime Databases, and visualized through a React.js dashboard. The system achieves a classification accuracy of 83.95 %, a macro F1-score of 84.7 %, and an average latency of 5 seconds, making it suitable for real-time applications. The proposed system offers a lightweight, synchronized, and deployable solution for emotion recognition, with potential applications in healthcare, elder care, and emotion aware interactive systems.**

***Index Terms*—Emotion Recognition, Multimodal Analysis, Real-Time Processing, Physiological Monitoring, Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Random Forest Classifier, Integration,**

I. Introduction

The increasing role of artificial intelligence in fields such as healthcare, smart wearable, and interactive computing has created new opportunities for systems that can detect and respond to human emotions [1]. While conventional methods focus on analysing a single input like facial expressions, voice tone, or physiological signals these unimodal systems often face limitations in dynamic environments. Issues like background noise, inconsistent lighting, and individual variability can reduce their effectiveness in real-world settings [2], [3]. To address these challenges, recent research has turned towards multimodal emotion recognition, a technique that leverages multiple data sources to improve detection accuracy and reliability. By combining facial, vocal, and physiological signals, systems can gain a complete understanding of a person's emotional state [3]. This paper presents a real time multimodal emotion recognition framework that integrates three key modalities: physiological signals (heart rate, SpO_2, body temperature), facial expressions, and speech tone. The system is designed to enhance accuracy and robustness by using signal specific preprocessing and machine learning models tailored to each modality. A fusion strategy is employed to combine the predictions from the individual models. Data is acquired through real time sensor streams from an ESP32-based Arduino microcontroller and publicly available datasets (CREMA-D, SAVEE, TESS, FER-2013) [4], [5], [6], [7]. This data is processed through a lightweight architecture utilizing Firebase Realtime Database for data acquisition, synchronization, and visualization, with a React.js dashboard for real-time monitoring and visualization of predictions [8]. Early evaluations of individual modalities reveal significant disparities in their predictive strength. The physiological signal model for example, achieves an classification accuracy of 99.23 %, demonstrating its robustness and reliability. In contrast, the facial and speech models report lower accuracies of 64 % and 62 %, respectively due to challenges such as occlusion, lighting variations, and background noise. These findings underscore the importance of multimodal fusion in overcoming the limitations of individual modalities and improving overall system performance [2], [9], [10]. The major contributions of this work are as follow:

- Multimodal Fusion Strategy: The integration of complementary features from sensors (MAX30102 for heart rate and SpO_2, DS18B20 for body temperature), facial landmarks (processed through OpenCV and the FER-2013 dataset), and MFCC-based speech features using a late fusion meta classifier.
- Lightweight Deployment Framework: A real time emotion recognition system with a prediction latency of 5 seconds, visualized through a custom web dashboard built using React.js and synchronized with fire base real time database for continuous updates.
- Expanded Dataset & Generalization: A combination of real time physiological data and publicly available annotated datasets (FER-2013, CREMA-D, SAVEE, TESS), augmented with techniques like normalization and feature extraction to enhance model robustness.

- Preliminary experiments show that the proposed system achieves an overall testing accuracy of 83.95 % and an F1 score of 84.7 % outperforming traditional unimodal systems.

The structure of this paper is as follows: Section II reviews related work, Section III outlines the design methodology, Section IV presents results and evaluations, and Section V concludes with directions for future work.

II. Literature Survey

Emotion recognition has gained significant attraction in recent years as a crucial component for intelligent systems used in various sectors such as healthcare, smart environments, and mental wellness platforms. As real time interactions become more integrated, unimodal approaches have become inadequate in dynamic, real world settings. This section highlights the development of multimodal systems and examines current methods in emotion recognition.

A. Detection of Unimodal Emotions

Single modality was the focus of early emotion recognition efforts. Early CNN models, such as VGG16, were able to detect visual emotion cues through facial expression recognition using datasets like FER2013 [4] and methods like Local Binary Patterns (LBP). However, when these models encountered difficulties like occlusion, shifting camera angles or changes in lighting, their performance significantly declined. For training models, datasets such as TESS [5], SAVEE [6], and CREMA-D [7] have been thoroughly investigated in literature. Emotion classification using speech features like MFCCs, pitch, and chroma has been widely applied in the audio domain [12]. For example, L.Cohn achieved approximately 65 % accuracy on CREMA-D by combining MFCCs with different classifiers. However, these models were sensitive to noise and microphone variability limiting their effectiveness in practical settings. Physiological signal based approaches, on the other hand, have shown more consistency in controlled conditions. Parameters such as ECG, heart rate, and SpO_2 have been explored using techniques like wavelet transforms and neural networks [13]. These physiological signals, although rich in emotion linked data are often subject to individual differences and lack contextual cues. Despite achieving solid baseline performance, unimodal models in physiological signal based emotion detection are limited by environmental factors such as sensor noise and variability.

1) Multimodal Fusion Techniques: To overcome the limitations of unimodal approaches, researchers have explored multimodal emotion recognition systems. Zadeh et al. [9] proposed a Tensor Fusion Network (TFN) that combines visual, vocal, and physiological features for emotion detection. However, TFN systems require large and synchronized datasets which are rare in emotion research. Ayubi et al. [10] used wearable sensor data to track emotions for mental health applications, but this system lacked additional behavioural inputs like speech or facial expressions, restricting its emotional resolution. Tripathi et al. [11] employed transformer based fusion techniques to improve performance across text, speech, and visual cues but encountered computational overheads. Wang et. al. [14] proposed a deep learning based framework that integrates facial and speech features using feature-level fusion, enabling real-time emotion recognition. While effective in improving classification accuracy, the system did not incorporate physiological signals, limiting its coverage of internal emotional states. Similarly, Ghosal et al. [15] introduced the CosMIC dataset, focusing on audio, video, and physiological fusion. Although insightful, their system lacked a real time deployment framework limiting its practical applicability. Many systems continue to focus on decision level fusion, testing each modality separately or without synchronized fusion strategies. This has prompted researchers to improve reliability using ensemble learning methods although full integration of modalities remains a challenge.

B. Deep Learning Based Advances

Recent developments in CNNs, Long Short Term Memory networks (LSTMs), and transformers have revolutionized emotion recognition by enabling automatic feature extraction from raw data streams. For example, models trained on speech features had accuracy rates between 65 - 70 %, depending on noise levels [2], while visual models trained on datasets like FER2013 via transfer learning achieved 70-75 % accuracy [3].

Transformer based architectures have shown potential for cross modal reasoning as demonstrated by Tripathi et al. [11], Kumar et al. achieved over 92 % accuracy on ECG and heart rate data using neural networks under controlled conditions [16]. although these models require extensive training and are often lacking in interpretability. Despite these advancements, most emotion recognition systems still rely on single-modality pipelines, which reduces their adaptability to real-time emotion monitoring scenarios. A significant gap in current research is the integration of facial, physiological, and vocal data in real-time systems. Effective cross-modal fusion for emotion recognition is accomplished by very few frameworks, especially when low latency and real-time responses are desired in real-world applications. Several recurrent issues can be found in the existing literature:

- Dataset Limitations: The generalisation of models is impacted by the scale or emotional diversity limitations of many emotion datasets, including SAVEE [6] and CREMA-D [7].
- Model Diversity: Single modality networks, like LSTMs for speech emotion detection [2] or CNNs for facial emotion recognition [12] are still the foundation of many emotion recognition systems. The synergy between various modalities is lost in these systems.
- Adaptability: A lot of current systems rely on manually created features and rule based models (LBP) [12], MFCCs [13], that are not very effective in dynamic and real world settings.

Prior multimodal emotion recognition studies often lack synchronization, physiological inputs, real-time deployment, or effective fusion. Proposed work addresses these gaps using

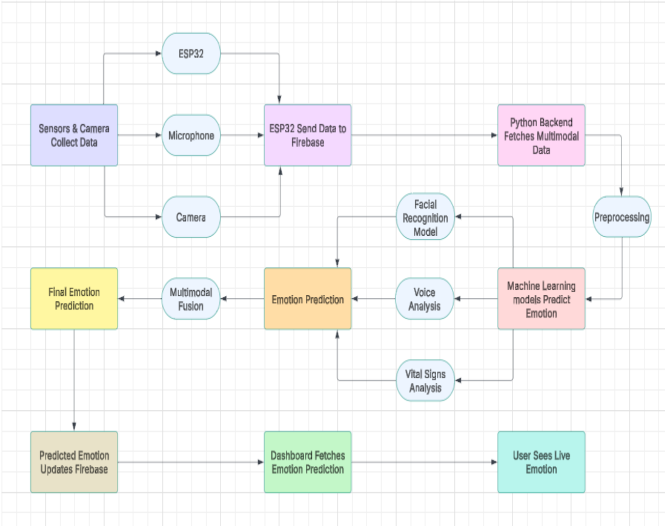

Figure 1. System architecture for real-time multimodal emotion detection

an ESP32 Firebase pipeline with a weighted fusion strategy (60 % physiological, 20 % facial, 20 % vocal) that ensures robustness, low latency, and adaptability.

III. Proposed Methodology

This work aims to develop a real time multimodal emotion detection system by integrating physiological, vocal, and facial data streams. The proposed framework employs an ensemble of pretrained models, leveraging sensor data, audio features, and image data for accurate emotion prediction. The system architecture utilizes Firebase Realtime Database for effective data synchronization and React.js for real-time visualization of emotion predictions.

A. *System Overview*

The proposed methodology consists of several interdependent stages, each designed to process and predict emotions based on multimodal data streams. These stages ensure that data is acquired, processed, synchronized, and fused in real time for reliable emotion detection. Figure 1 shows that the system architecture integrates data from physiological sensors, vocal datasets, and real time webcam input. These data streams are synchronized via Firebase and visualized in real time using a React.js dashboard.

- Physiological Data: Heart rate, SpO2, and body temperature are recorded using an ESP32 microcontroller with MAX30102 (heart rate, SpO2) and DS18B20 (temperature) sensors. Data is collected at 1 Hz and uploaded to Firebase in real time.The ESP32 packages this data into a JSON object with a timestamp and transmits it via an HTTPS POST request to the Firebase Realtime Database.
- Vocal Data: Emotional speech features (RMS energy, pitch, spectral centroid, MFCCs) are extracted from CREMA-D, TESS, and SAVEE datasets using Librosa, and uploaded to Firebase every 5 seconds.

TABLE I
Comparison of Data Acquisition Modalities

Datatype	**Sensor/Dataset**	**Features Extracted**	**Sampling Rate**
Physiology	MAX30102, DS18B20	Heart Rate, SpO_2, Body Temp	1 Hz
Vocal	CREMA-D, TESS, SAVEE	RMS Energy, Pitch, MFCCs, Spectral Centroid	5 sec
Facial	Webcam, FER-2013	Facial Expressions	5 sec

- Facial Data: Real-time webcam feeds are processed using OpenCV for face detection and normalization. Offline training uses FER-2013. Processed images are uploaded to Firebase every 5 seconds.

As summarized in Table I, the system collects data from three key modalities using different sensors and datasets at defined intervals. For vocal analysis, libraries such as Librosa are employed to extract features before feeding them into a model. For facial analysis, OpenCV is used to detect and crop faces from the webcam feed for processing. The following steps are involved in the synchronization process:

- Getting the most recent entries from the Firebase database for each modality
- Data aggregation and timestamping to guarantee proper temporal alignment.
- Gathering and forwarding the combined data to a fourth Firebase database, where all modalities are stored for future examination.

Accurate emotion detection depends on the data from each modality remaining aligned which is ensured by this process. The hardware includes an ESP32 microcontroller, a MAX30102 sensor, a DS18B20 temperature sensor, a USB webcam, and a microphone. The MAX30102 sensor is connected to the ESP32's I^2C pins (SCL to GPIO 22 and SDA to GPIO 21), while the DS18B20 sensor is connected via a one-wire protocol, with its data pin (DQ) connected to a digital pin (GPIO 4) and a 4.7k Ohm pull-up resistor between DQ and VDD. The USB webcam and microphone are connected directly to a PC for data acquisition.

B. *Preparing Data and Extracting Features*

For each data modality to be suitable for feature extraction and emotion classification, particular preprocessing procedures must be followed. Data collected via sensors (heart rate, SpO2, body temperature), then the following steps are implemented:

- Data Normalization: Raw data is normalized to a standard range of [0, 1] to ensure consistency and improve model training.
- Outlier Removal: Extreme outliers are identified using standard deviation-based filtering techniques and removed to ensure clean input data.
- Data Smoothing: A moving average filter is applied to smooth out fluctuations in the physiological signals and make the data more representative of trends.

These pre-processed data features are then ready for use by the Random Forest model for emotion classification. Audio features (RMS, pitch, MFCCs, spectral centroid) are:

- Feature Normalization: The extracted features (RMS energy, pitch, spectral centroid, MFCCs) are normalized to standard ranges to ensure that the model training process is efficient.
- Dimensionality Reduction: Principal Component Analysis (PCA) is applied to reduce the dimensionality of the feature set while preserving essential information. This step helps the LSTM model process data more effectively.
- Noise Filtering: Audio signals are filtered to remove background noise using spectral gating methods to ensure clarity in emotion-related vocal features.

These features are then used as input to the LSTM-based Recurrent Neural Network (RNN) for emotion prediction. The facial images undergo preprocessing steps to prepare them for input into the CNN model:

- Face Detection: Faces are detected using the Haar Cascade classifier from OpenCV. Any extraneous information (e.g., background) is discarded.
- Resizing and Normalization: Detected faces are resized to 200×200 pixels and normalized to the range [0, 1] to standardize inputs for the CNN.
- Augmentation: To ensure robustness, training data undergoes augmentation techniques such as horizontal flips, rotations, and changes in brightness. This simulates real-world variations in facial expressions improving model generalization.

These preprocessed images are then input to the CNN for facial emotion classification.

C. Emotion Prediction

Each modality is classified using a dedicated model:

Model 1: Facial Emotion Detection (CNN Model)

The CNN model is structured as follows:

- The input is a 200×200 pixel image representing a face.
- The network consists of multiple convolutional layers followed by pooling layers that gradually abstract features from low-level textures to high-level facial patterns.
- The output of the CNN is a vector of emotion probabilities, one for each emotion category.

Model 2: Vocal Emotion Detection (LSTM Model)

The model architecture includes:

- The input consists of a sequence of vocal features over time.
- LSTM layers are used to capture the sequential nature of the data.
- The final output is a probability distribution over the possible emotions based on the vocal features.

Model 3: Physiological Emotion Detection (Random Forest Model)

Each modality is classified using a dedicated model:

- CNN for facial data (trained on FER-2013)
- LSTM for vocal features (captures temporal cues)
- Random Forest for physiological signals (achieves 99.2 % accuracy on test data)

TABLE II
COMPARISON OF DIFFERENT FUSION WEIGHTS

Fusion Weights (Physio–Facial–Vocal)	Accuracy (%)	F1-score (%)
33–33–33 (Equal)	78.42	78.90
50–25–25	81.21	81.55
60–20–20 (Proposed)	83.95	84.70
70–15–15	82.33	82.50

D. Fusion Strategy

The predictions from all three models are combined using a weighted fusion strategy:

- 60 % weight is given to physiological data (due to its strong correlation with emotional states).
- 20 % weight is assigned to facial data (due to its relevance in recognizing facial expressions).
- 20 % weight is assigned to vocal data (which provides crucial emotional cues through speech patterns).

As compared in Table II, the proposed configuration (60-20-20), equal weighting (33-33-33), moderately physiologically biased weighting (50-25-25), and a highly physiologically dominant weighting (70-15-15) were among the alternative configurations tested to validate the selection of fusion weights.The findings demonstrated that while placing more emphasis on physiological data enhanced performance, equal weighting resulted in the lowest accuracy (78.42 %) and F1 score (78.90 %). The suggested weighting of 60 % physiological, 20 % facial, and 20 % vocal produced the best accuracy (83.95 %) and F1 score (84.70 %), demonstrating how well it balanced the contributions of each modality. Performance dropped slightly to 82.33 % accuracy and 82.50 % F1 score when the physiological weight was increased to 70 %, suggesting that an excessive dependence on one modality can weaken overall robustness. The fused prediction is stored in Firebase, where it is available for real time visualization.

IV. Results and Discussion

This section validates the efficacy of the proposed multimodal emotion detection system by evaluating each model, analyzing performance, visualizing training behavior, and comparing fused and unimodal approaches. The system operates by first collecting a constant stream of physiological data (heart rate, SpO_2, temp), facial, and vocal. This raw data is then sent to a cloud server. On the server, three separate machine learning models analyze each data type individually. Finally, the core innovation of the system, a weighted fusion strategy, combines individual predictions into a single highly accurate emotion prediction, which is then displayed on a real-time dashboard.

A. Modality-wise Emotion Recognition Performance

Figure 2 depicts circuit connections between the sensors (MAX30102, DS18B20) and the ESP32 microcontroller. The Random Forest model trained on physiological data (Heart

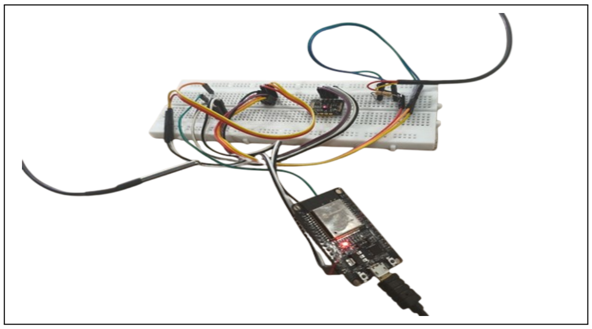

Figure 2. Circuit Implementation/Circuit Connection

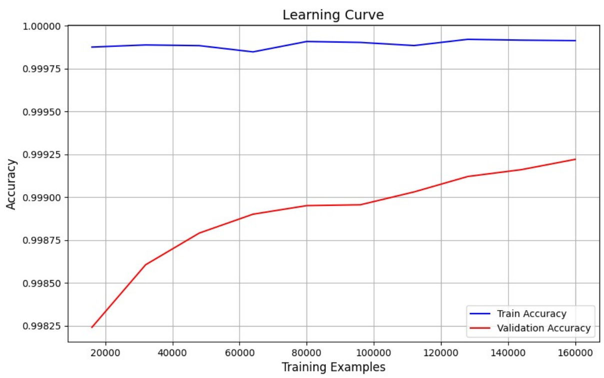

Figure 3. Training accuracy for physiological model

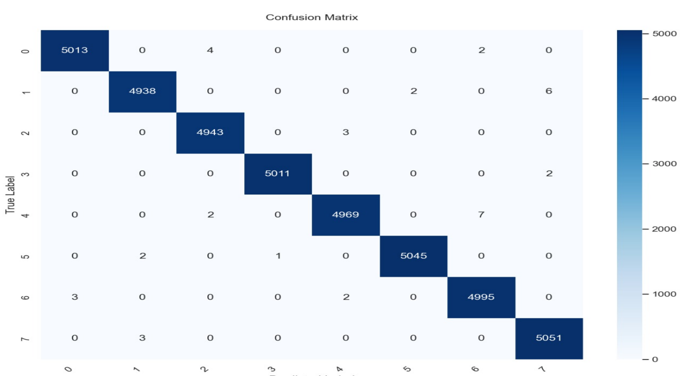

Figure 4. Confusion matrix for physiological emotion classification.

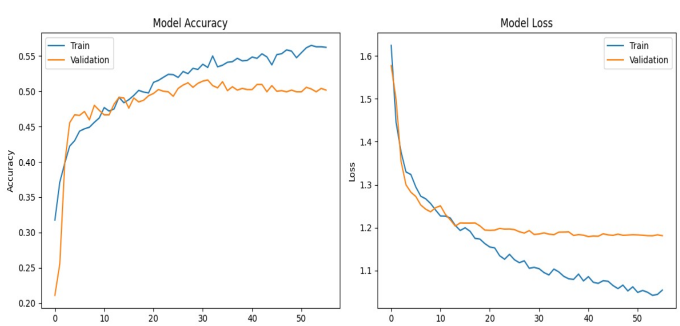

Figure 5. Training and validation accuracy/loss for vocal modality

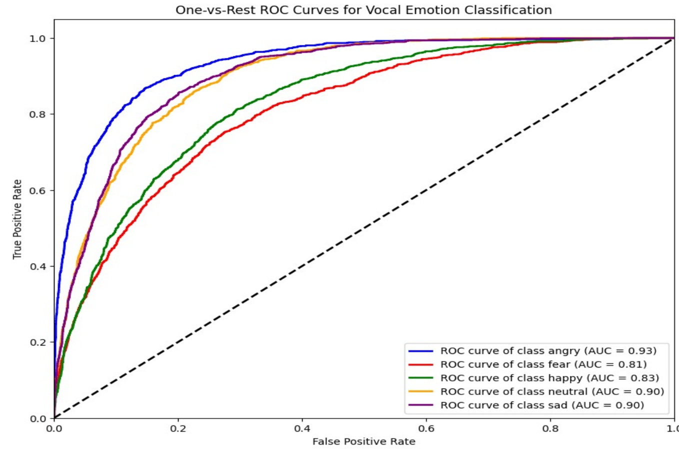

Figure 6. ROC Curve for Vocal Modality

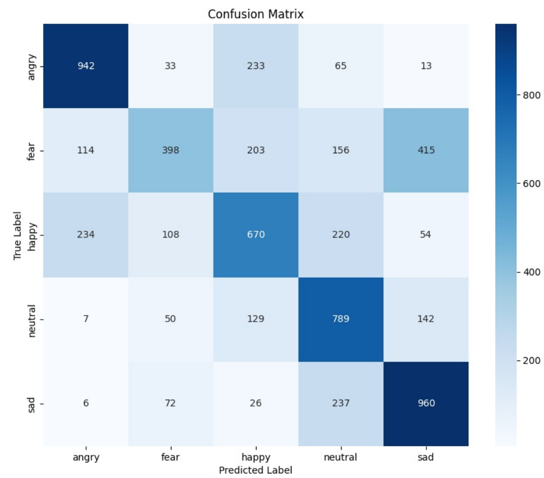

Figure 7. Confusion matrix for vocal emotion classification using LSTM

Rate, SpO_2, and Temperature) achieved high classification accuracy. As shown in Figure 3, the model converges quickly with minimal overfitting, indicating efficient learning from biometric signals. The confusion matrix in Figure 4 simplifies classification into six core emotions. Misclassifications were observed between Angry and Sad emotions likely due to similar HR and SpO_2 fluctuations under stress. The vocal emotion classifier, implemented using an LSTM model, demonstrated moderate performance. Figure 5 shows training and validation accuracy/loss curves. While training was stable, the model converged slower than physiological one, likely due to MFCC distortions from background noise. Figure 6 shows the One-vs-Rest ROC curves for vocal emotion classification. The model achieves strong performance with AUC scores of 0.93 (angry), 0.90 (neutral and sad), 0.83 (happy), and 0.81 (fear), demonstrating effective class discrimination. The confusion matrix in Figure 7 summarizes the model's classification performance across emotion classes (happy, sad, fear, angry,

Figure 8. CNN model accuracy/loss curves for facial modality

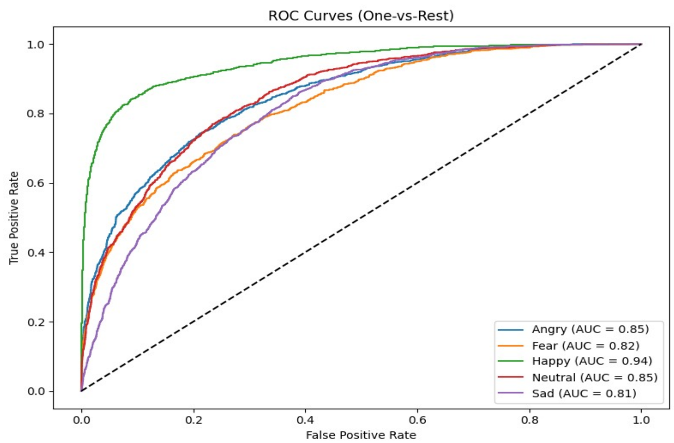

Figure 9. ROC Curve for Facial Modality

neutral). The CNN model trained on the FER-2013 dataset (and fine-tuned using webcam data) is visualized in Figure 8. Although training accuracy increased steadily, the validation curve plateaued early, indicating signs of overfitting. The final accuracy was approximately 64 %, significantly lower than the physiological performance.

Figure 9 shows that the facial emotion recognition performs strongest for happiness (AUC=0.94) and weakest for sadness (AUC=0.81), with anger/neutral at 0.85. Figure 10 shows that most confusion occurs between Sad and Neutral classes, indicating limitations in FER-2013 data granularity.

B. *Fusion-Based Emotion Prediction*

The three modalities were combined using a weighted fusion strategy: 60 % physiological, 20 % vocal, and 20 % facial. This

TABLE III
MODALITY-WISE PERFORMANCE METRICS

Modality	Accuracy	Precision	Recall	F1-Score
Facial (CNN)	60.18 %	61.02 %	60.18 %	60.36 %
Vocal (LSTM)	62.00 %	60.00 %	60.00 %	59.00 %
Physio (RF)	99.93 %	99.93 %	99.93 %	99.93 %
Fusion	83.95 %	84.80 %	84.90 %	84.70 %

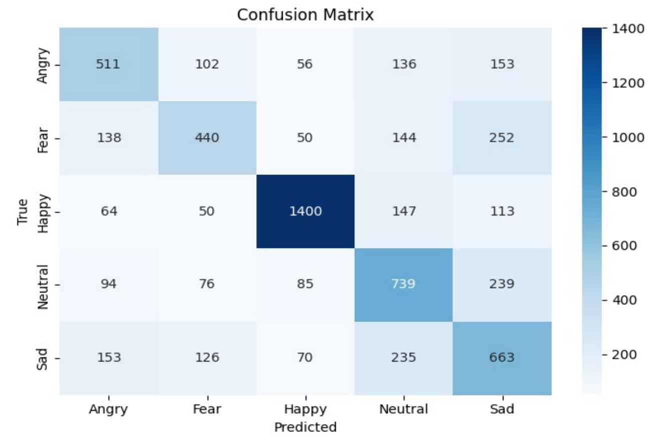

Figure 10. Confusion matrix for facial emotion classification using LSTM.

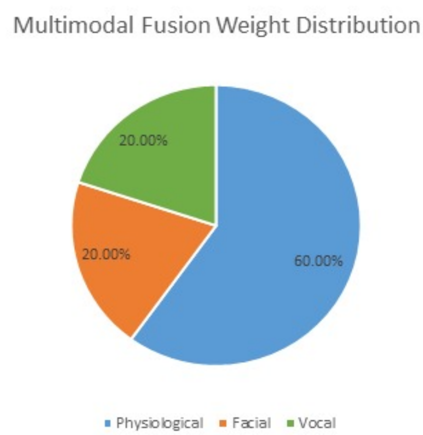

Figure 11. Final Emotion Prediction combining modalities with pie chart of fusion weights.

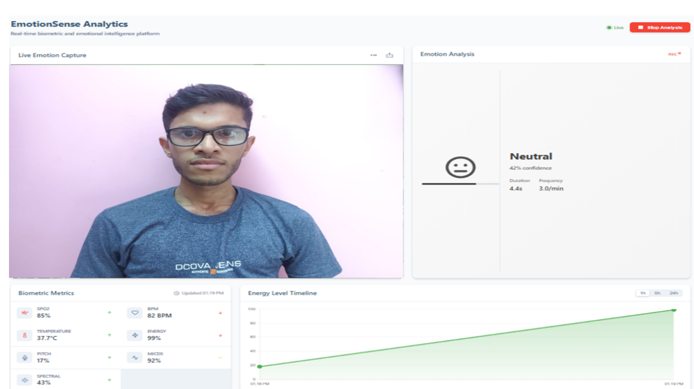

Figure 12. Real-time analytics dashboard showing emotion predictions, sensor readings, and confidence scores

TABLE IV
COMPARISON WITH PRIOR WORK

Reference	Dataset(s)	Features	Techniques	Accuracy (%)
[2]	IEMOCAP, DEAP, RECOLA	Facial expressions, speech prosody, physiological signals (HR, EEG)	CNN, LSTM, Fusion Techniques	87.0
[10]	CMU-MOSEI	Facial, vocal, textual	Transformer-based Fusion Model	91.2
[17]	IEMOCAP, MELD, MOSEI	Audio, visual, textual data	Modality fusion with graph contrastive learning	State-of-the-art
[18]	RoBERTa, Wav2Vec2, FacialNet	Text, speech, facial expressions, video	CNN + Transformer	66.36 (emotion), 72.15 (sentiment)
[19]	FER-2013, RAVDESS, ETD	Audio, video, text	CNNs for each modality, decision-level fusion	80.0
Proposed Work	CREMA-D, SAVEE, TESS, FER-2013, Human Vital Sign Dataset	Physiological (BPM, SpO_2, Body Temp.), vocal, facial	Random Forest, CNN, LSTM	83.95

weighting was based on individual performance and feature.

Table III shows that unlike existing works that rely on data sets like IEMOCAP, MOSEI, or DEAP with EEG, text, or prosody features, our system uses real-time physiological signals (HR, SpO_2, Temp), webcam facial images (FER-2013), and vocal data (MFCCs of CREMA-D, SAVEE, TESS). The paper [2] achieves a high accuracy of 87 % with EEG, but that is an invasive and non-practical modality while this work achieves a competitive 83.95 % accuracy with accessible sensors, making it more suitable for real world applications. The another study [10] uses a textual modality, which is a significant constraint for non-verbal communication, and achieves 91.2 % accuracy. The robustness of this system's emotion detection without text is one of its strongest points. To achieve state of the art results, the work [17] employs complex graph contrastive learning; however, these techniques are frequently computationally costly and require offline analysis but our framework offers a concrete, deployable contribution by concentrating on a real-time, lightweight implementation with a personalized dashboard. Although the prior work [18] employs a multimodal approach as well, the accuracy is significantly lower whereas the proposed system's much higher accuracy of 83.95 % shows how well the model selections worked and how the fusion strategy outperformed their approach. The work [19] achieves 80 % accuracy using audio and video with a straightforward decision-level fusion. By including physiological modality and employing a weighted fusion strategy, my system outperforms this with 83.95 % accuracy. We employ modality-specific models- Random Forest for physiological, CNN for facial, and LSTM for vocal data- fused via a weighted strategy of 60-20-20 as visualized in Figure 11. As shown in Table IV, the system achieves an accuracy of 83.95 %, macro F1-score of 84.7 %, and 5-second latency, making it suitable for real time emotion recognition.

C. *Real-Time Deployment Results*

A working prototype of the Emotion Sense system was deployed and visualized using a React.js dashboard with Firebase. As shown in Figure 12, emotion prediction updates every 4.4 seconds. Sample values include Heart Rate (82 BPM), S (85 %), and a confidence score of 42 % reduced due to vocal interference in noisy environmets.

V. CONCLUSION AND FUTURE WORK

Using an ESP32 Firebase pipeline, we created a real-time multimodal emotion recognition system that combines vocal, facial, and physiological inputs. The system achieved low latency and 83.95% accuracy by using a weighted fusion strategy. Future work will concentrate on extending the range of physiological signals (such as EEG), integrating adaptive fusion weight tuning through reinforcement learning, and deploying the framework on embedded edge devices for standalone operation, even though the current system achieves robust multimodal emotion recognition with low latency. Furthermore, longitudinal research will be carried out to verify performance in various real world settings, including healthcare facilities and senior living facilities.

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